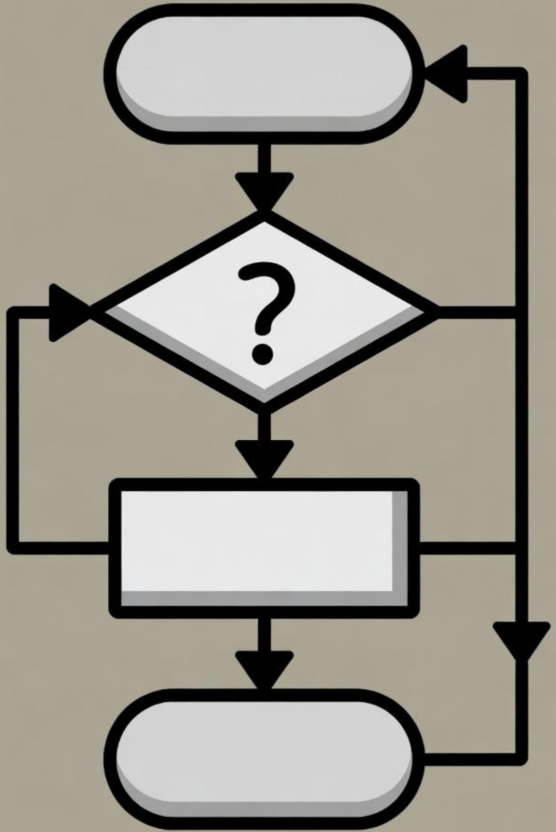

알고리즘 (Algorithm)

AI 신약

신현길

실습 환경

- Python
  - anaconda (miniconda)
  - jupyter notebook
- Terminal
  - Anaconda prompt

Miniconda Installer

Miniconda is a free minimal installer for conda. It is a small bootstrap version of Anaconda that includes only conda, Python, the packages they both depend on, and a small number of other useful packages (like pip, zlib, and a few others).

Download Miniconda Installer

초등학생 시절 배운 알고리즘

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\begin{array}{r}
5678 \\
\times\ 1234 \\
\hline
\end{array}
$$

계산 하는 방법?

정답: 7,006,652

정확한 곱셈 계산을 위해 필요한 물품?

분자구조 1개에서 분자 지문을 계산하는데 1초가 걸린다면…

계산이 오래 걸리는걸까?

덕성여자대학교 포트 × 덕성여자대학교 통합 × 덕성여대 웹메일-1 × 덕성여자대학교 e-C × ZINC × 국가슈퍼컴퓨팅센터 × 국가슈퍼컴퓨팅센터 × semi supervised lea ×

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ZINC20

Welcome to ZINC, a free database of commercially-available compounds for virtual screening. ZINC contains over 230 million purchasable compounds in ready-to-dock, 3D formats. ZINC also contains over 750 million purchasable compounds you can search for analogs in under a minute.

ZINC is provided by the Irwin and Shoichet Laboratories in the Department of Pharmaceutical Chemistry at the University of California, San Francisco (UCSF). We thank NIGMS for financial support (GM71896).

To cite ZINC, please reference: Irwin, Tang, Young, Dandarchuluun, Wong, Khurelbaatar, Moroz, Mayfield, Sayle, *J. Chem. Inf. Model 2020*, in press. https://pubs.acs.org/doi/10.1021/acs.jcim.0c00675. You may also wish to cite our previous papers: Sterling and Irwin, *J. Chem. Inf. Model, 2015* *http://pubs.acs.org/doi/abs/10.1021/acs.jcim.5b00559*. Irwin, Sterling, Mysinger, Bolstad and Coleman, *J. Chem. Inf. Model, 2012* DOI: 10.1021/ci3001277 or Irwin and Shoichet, *J. Chem. Inf. Model. 2005;45(1):177-82* PDF, DOI.

Getting Started

- Getting Started
- What's New
- About ZINC 20 Resources
- Current Status / In Progress
- Why are ZINC results "estimates"?

Explore Resources

Chemistry
Tranches, Substances, 3D Representations, Rings, Patterns
And More
Catalogs, Genes, ATC Codes

Ask Questions

You can use ZINC for **general** questions such as

- How many substances in current clinical trials have PAINS patterns? (150)
- How many natural products have names in ZINC and are not for sale? (9296) get them as SMILES, names and calculated logP
- How many endogenous human metabolites are there? (47319) and how many of these can I buy? (8271) How many are FDA approved drugs? (94)
- How many compounds known to aggregate are in current clinical trials? (60)
- How many epigenetic targets have compounds known? (53) and Which of these substances can I buy? (278)
- How many ligands are there for the NMDA 1 ion channel GRIN1? (662) and How many of these are for sale? (60)
- More...

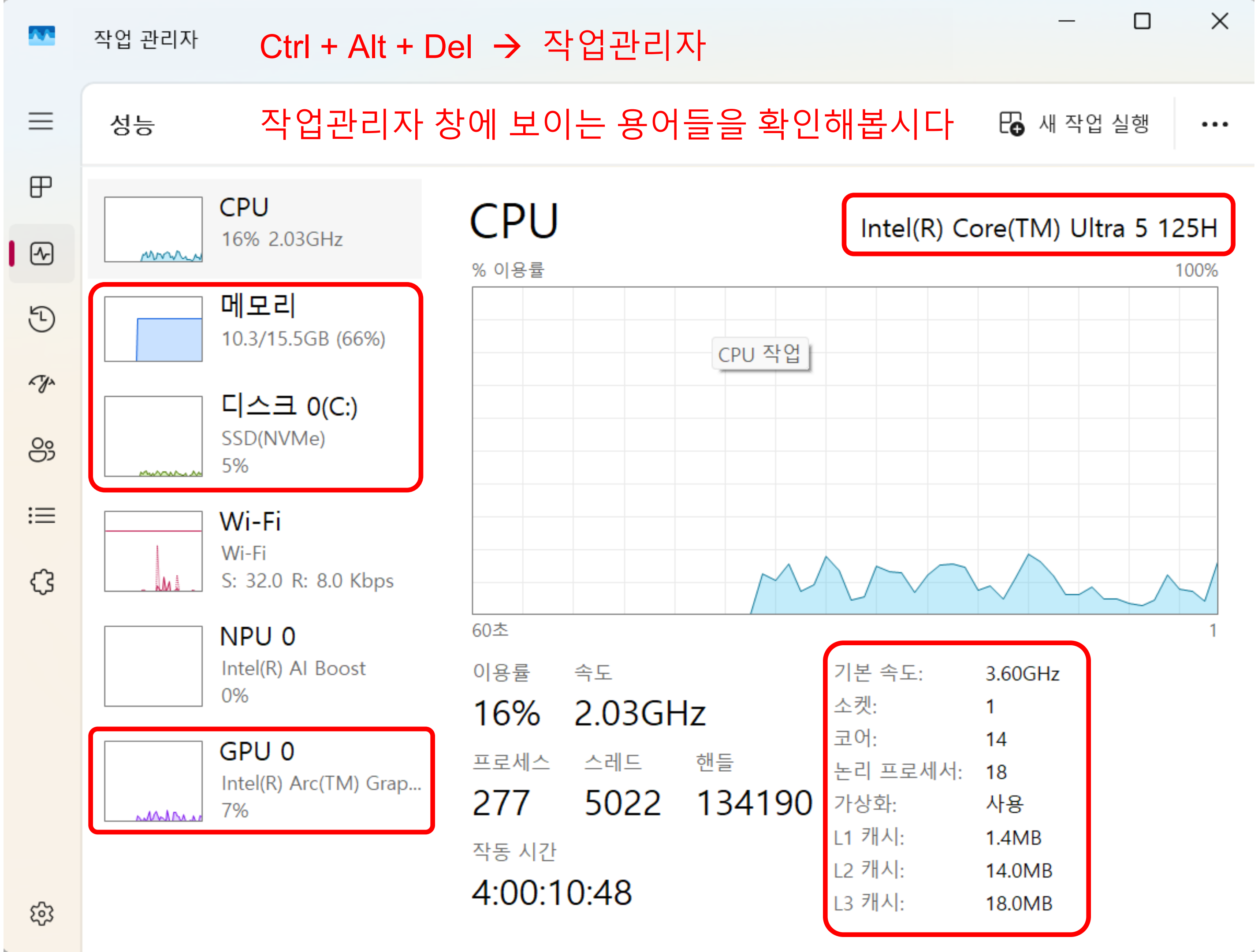
ZINC20 News

- ZINC20 has been released

Caveat Emptor: We do not guarantee the quality of any molecule for any purpose and take no responsibility for errors arising from the use of this database. ZINC is provided in the hope that it will be useful, but you must use it at your own risk.

알고리즘이란?

- 문제를 해결하는 방법
    - 문제에 대한 답을 구하는 방법은 매우 다양
    - 답변의 정확성과 **문제를 해결하는데 걸리는 시간**이 중요
    - 아무리 정확한 답변도 너무 오랜 시간이 걸리면 문제
- 문제 해결 과정에서 사용할 수 있는 계산 자원
    - CPU
    - Memory
    - Disk
    - GPU


작업 관리자
Ctrl + Alt + Del → 작업관리자
성능
작업관리자 창에 보이는 용어들을 확인해봅시다
새 작업 실행
CPU
16% 2.03GHz
메모리
10.3/15.5GB (66%)
디스크 0(C:)
SSD(NVMe)
5%
Wi-Fi
Wi-Fi
S: 32.0 R: 8.0 Kbps
NPU 0
Intel(R) AI Boost
0%
GPU 0
Intel(R) Arc(TM) Grap...
7%
CPU
Intel(R) Core(TM) Ultra 5 125H
% 이용률
100%
CPU 작업
60초
1
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16%
2.03GHz
프로세스
스레드
핸들
277
5022
134190
작동 시간
4:00:10:48
기본 속도: 3.60GHz
소켓: 1
코어: 14
논리 프로세서: 18
가상화: 사용
L1 캐시: 1.4MB
L2 캐시: 14.0MB
L3 캐시: 18.0MB

여러 명이 함께 힘을 합해서 더 빠르게 계산하는 방법?

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KSC (Korea Supercomputing Center)

https://www.ksc.re.kr/

가입은 학교 메일로! (duksung.ac.kr)

분자구조 1개에서 분자 지문을 계산하는데 1초가 걸린다면...

KSC CPU 서버를 이용해서 1초에 50만개씩 계산할 수 있다면, 8억개를 계산하는데 걸리는 시간은?

덕성여자대학교 포털 | 덕성여자대학교 통합 | 덕성여대 웹메일-1 | 덕성여자대학교 e-C | ZINC | 국가슈퍼컴퓨팅센터 | 국가슈퍼컴퓨팅센터 | semi supervised lea

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1,916 (Non-Empty) Tranches Selected (872,843,544 Substances)

AAAA AAAB AAAC AAAD AABA AABB AABC AABD AACA AACB AACC AACD AAEA AAEB AAEC AAED BAAA BAAB BAAC BAAD BABA BABB BABC BABD BACA BACB BACC BACD BAEA BAEB BAEC BAED CAAA CAAB CAAC CAAD CABA CABB CABC CABD CACA CACB CACC CACD CAEA CAEB CAEC CAED DAAA DAAB DAAC DAAD DABA DABB DABC DABD DACA DACB DACC DACD DAEA DAEB DAEC DAED EAAA EAAB EAAC EAAD EABA EABB EABC EABD EACA

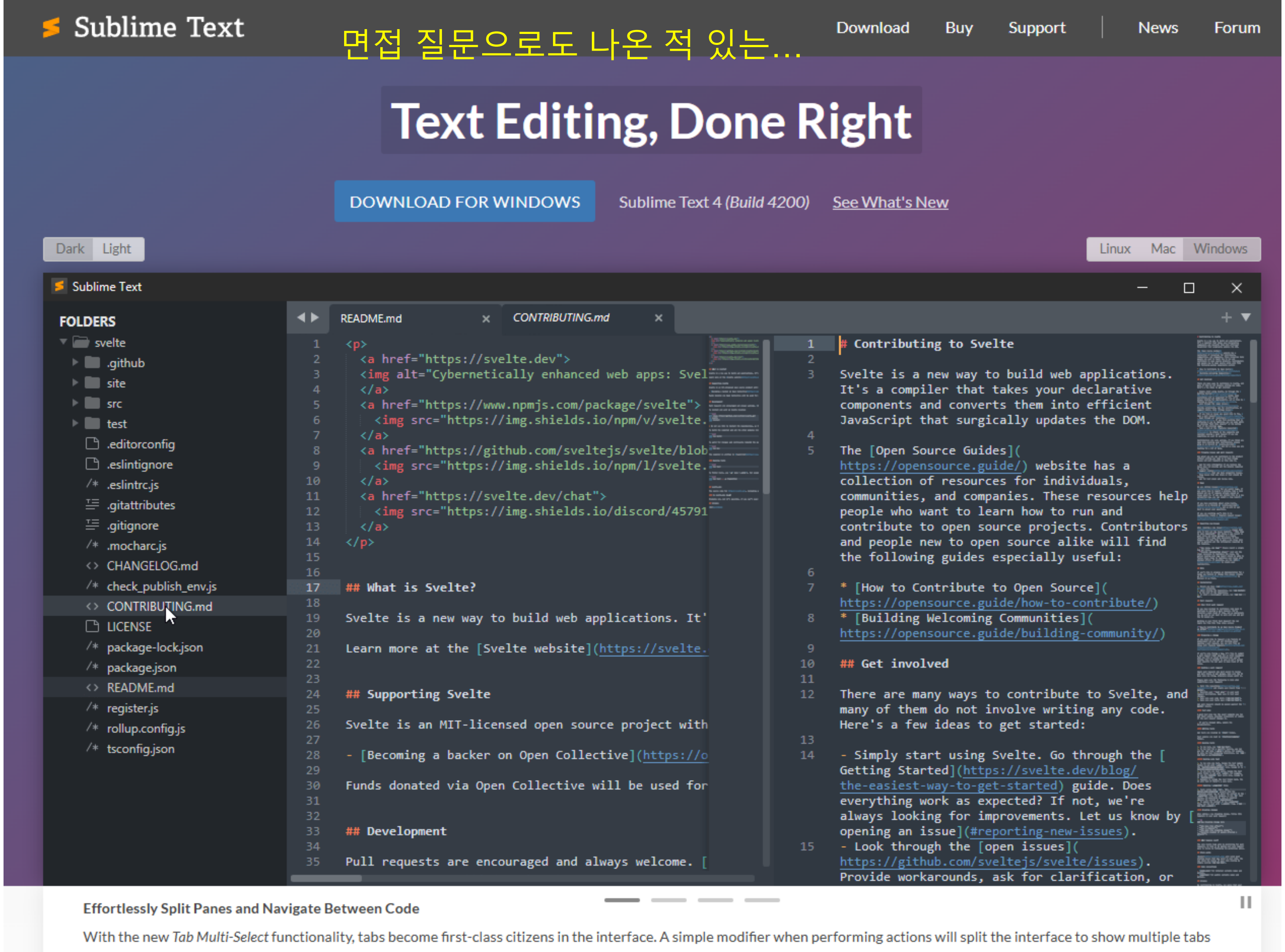
Only If Modified Since

연도-월-일

Tab-Delimited (*.txt) | Raw URLs | Download

LogP (up to)	200	250	300						450	500	>500	Totals, by LogP
-1	28,823	172,492							92,390	77,937	7,897	5,582,708
0	144,952	933,153							570,371	507,292	4,737	27,399,373
1	378,195	2,881,833							2,647,157	2,412,007	9,514	91,933,357
2	486,334	4,591,795							8,090,798	7,679,805	21,862	193,454,132
2.5	173,268	2,143,741							7,183,481	6,968,353	19,147	126,326,561
3	93,389	1,576,170							9,339,902	9,097,041	24,917	128,408,335
3.5	37,751	933,238	7,912,551	12,468,392	27,344,261	18,657,854	16,443,198	11,750,376	10,740,422	10,658,718	31,899	116,978,660
4	9,297	371,018	4,326,589	6,458,909	10,461,106	12,996,329	14,287,881	11,641,267	10,847,916	10,954,131	37,915	82,392,358
4.5	1,038	86,858	1,810,826	3,448,745	6,348,381	8,824,160	2,087,870	9,902,447	9,440,116	9,765,320	41,819	51,757,580
5	172	13,518	534,506	1,400,788	3,156,627	4,976,069	6,445,496	6,991,949	6,934,172	7,261,478	45,578	37,760,353
>5	45	1,150	22,281	100,633	367,407	907,349	1,637,877	2,150,056	2,525,726	2,891,738	245,865	10,850,127
Totals, by Weight	1,353,264	13,704,966	77,732,975	111,242,202	232,552,145	127,652,778	97,231,027	74,196,766	68,412,451	68,273,820	491,150	**872,843,544** Substances **1.9K** Tranches

면접 질문으로도 나온 적 있는...



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다운로드 버튼 X

smiles	zinc_id	inchikey	mwt	logp	reactive	purchasable	tranche_name	features
CO[C@H]1OC[C@@H](O)[C@H](O)[C@H]1O	4371221	ZBDGHWFPLXXWRD-MOJAZDJTSA-N	164.157	-1.928	0	50	AAAA	
C[S@@](=O)CC(N)=O	34310585	TTXPTDCYRCXFHQ-ZETCQYMHSA-N	121.161	-1.150	0	50	AAAA	
NC(=O)N[C@@H]1NC(=O)NC1=O	1843030	POJWUDADGALRAB-PVQJCKRUSA-N	158.117	-2.180	0	50	AAAA	endogenous,world
O=c1[nH]c(O)c(O)c(=O)[nH]1	1847417	YFQOVSGFCVQZSW-UHFFFAOYSA-N	144.086	-1.526	0	50	AAAA	biogenic,in-vitro
NC(=O)CN1CCC(N)CC1	9256947	JWRVCVWALQWYIJ-UHFFFAOYSA-N	157.217	-1.105	0	50	AAAA	in-vitro
CNC(=O)c1n[nH]c(N)n1	19844301	AFLLHPQFSIEJFI-UHFFFAOYSA-N	141.134	-1.254	0	50	AAAA	
NCc1cc(=O)[nH]c(O)n1	26507110	UMEHPGXESZPRFZ-UHFFFAOYSA-N	141.130	-1.066	0	50	AAAA	in-vitro
CS(=O)(=O)CCCNN	63014624	ZQJQZDHBEXECJD-UHFFFAOYSA-N	152.219	-1.116	0	50	AAAA	
NC(=O)[C@H]1[C@H]2C=C[C@@H](O2)[C@@H]1N	242677143	TTWXSENPKHNTJG-ZXXMMSQZSA-N	154.169	-1.248	0	50	AAAA	
N[C@@H](CO)C(=O)NCCO	104591586	SNJHTJUWWOTLAG-BYPYZUCNSA-N	148.162	-2.586	0	50	AAAA	
CNS(=O)(=O)C[C@@H]1CNCCO1	238310355	DRWMVYSSBGMOSK-LURJTMIESA-N	194.256	-1.476	0	50	AAAA	
O=c1[nH]cc(N2CCOCC2)c(=O)[nH]1	55860	SIVJYKAWVLZRSX-UHFFFAOYSA-N	197.194	-1.100	0	50	AAAA	in-vitro
NCCOCCOCCN	1651868	IWBOPFCKHIJFMS-UHFFFAOYSA-N	148.206	-1.063	0	50	AAAA	in-vitro
CS(=O)(=O)CCS(N)(=O)=O	44594710	MASSUGWMWUCANP-UHFFFAOYSA-N	187.242	-1.681	0	50	AAAA	in-vitro
NS(=O)(=O)NC1CCOCC1	160854167	DRBVAWQDPGGWIH-UHFFFAOYSA-N	180.229	-1.042	0	50	AAAA	
CN1C(=O)NC(=O)C1=O	1532166	MJJCZWPYKMGQHU-UHFFFAOYSA-N	128.087	-1.305	0	50	AAAA	biogenic,in-vitro
COC(=O)[C@H](N)CC(N)=O	32098738	XWBUDPXCXXQEOU-GSVOUGTGSA-N	146.146	-1.638	0	50	AAAA	biogenic,in-vitro
NCCS(=O)(=O)CCO	38251012	XZILNQFHWDEPHG-UHFFFAOYSA-N	153.203	-1.648	0	50	AAAA	
O=C1N[C@@H]2NC(=O)N(CCO)[C@H]2N1	2251996	PKNZKKFFNYKLBG-QWWZWVQMSA-N	186.171	-2.031	0	50	AAAA	biogenic
O=C1O[C@@H]2C[C@@]1(O)C[C@H](O)[C@@H]2O	71786849	QPJRIFFWEBJVFN-RULRNXASSA-N	174.152	-1.842	0	50	AAAA	
O=C(O)C(=O)C[C@@H](O)[C@@H](O)CO	1532571	WPAMZTWLKIDIOP-WUJLRWPWSA-N	178.140	-2.256	0	50	AAAA	in-vitro,metabolite
CC(=O)NCCC(=O)NCCO	35100900	GSMTYNSCEYOUIG-UHFFFAOYSA-N	174.200	-1.379	0	50	AAAA	
Cn1nc(CN)cc1C(N)=O	39064903	WMUKXUJGJVOHRA-UHFFFAOYSA-N	154.173	-1.022	0	50	AAAA	
CNS(=O)(=O)NCCCN	160168213	BLCRPDVOORWVQR-UHFFFAOYSA-N	167.234	-1.611	0	50	AAAA	
O=C(O)C12CNCC1CNC2	306937672	IWHRKKRPBCVDRT-UHFFFAOYSA-N	156.185	-1.120	0	50	AAAA	
NCCc1nnc2c(=O)[nH]ccn12	409423317	PTNKDQUOYZTIJI-UHFFFAOYSA-N	179.183	-1.081	0	50	AAAA	
O=C(CO)[C@@H](O)[C@@H](O)CO	902219	ZAQJHHRNXZUBTE-UCORVYFPSA-N	150.130	-2.738	0	50	AAAA	biogenic,in-vivo
Cn1ncc2c(=O)[nH]c(NN)nc21	16969122	HAFKXNVWQWJQFI-UHFFFAOYSA-N	180.171	-1.058	0	50	AAAA	
O=C1NCCN[C@]1(O)C(F)(F)F	4204973	QCMGTMICABTCGL-SCSAIBSYSA-N	184.117	-1.043	0	50	AAAA	
NNc1n[nH]c(=S)n1N	12958520	RDIMQHBOTMWMJA-UHFFFAOYSA-N	146.179	-1.060	0	50	AAAA	in-vitro
Nc1nc(N)n2c(N)nnc2n1	16889989	KMZCSSCUBKHIJS-UHFFFAOYSA-N	166.148	-1.734	0	50	AAAA	
N[C@@H]1CNC[C@H]1O	34413647	ZCZJKVCIYONNJR-QWWZWVQMSA-N	102.137	-1.722	0	50	AAAA	
COC[C@H](N)CN	34426550	LNPOMYDQIBCEEO-SCSAIBSYSA-N	104.153	-1.081	0	50	AAAA	in-vitro
CNCc1nnn[nH]1	42191848	VRVRXSMTHQDPMY-UHFFFAOYSA-N	113.124	-1.081	0	50	AAAA	in-vitro
O=C(O)CNC(=O)CC(=O)O	45285524	IRELCDFUNQUKLG-UHFFFAOYSA-N	161.113	-1.338	0	50	AAAA	
COC(=O)[C@@H](O)[C@H](O)C(=O)O	1682447	GBJFSZCDZHSAOP-HRFVKAFMSA-N	164.113	-2.034	0	50	AAAA	in-vitro
CCNC(CO)CO	73836680	GWMADSUQNXLNCF-UHFFFAOYSA-N	119.164	-1.051	0	50	AAAA	
CN1CCOC(CO)(CO)C1=O	96024984	SBMKJZXVBJPLCN-UHFFFAOYSA-N	175.184	-1.802	0	50	AAAA	in-vitro
O=C(CO)[C@H](O)CO	901232	UQPHVQVXLPRNCX-GSVOUGTGSA-N	120.104	-2.099	0	50	AAAA	biogenic,in-vitro
N[C@@H](CCOP(=O)(O)O)C(=O)O	1532852	FXDNYOANAXWZHG-VKHMYHEASA-N	199.099	-1.102	0	50	AAAA	endogenous,in-man
CC(=O)N[C@@H](O)[C@H](O)NC(C)=O	3668823	VPSZYLKVQHCMJD-WDSKDSINSA-N	176.172	-2.105	0	50	AAAA	
O[C@@H]1[C@@H](O)[C@H](O)C=C[C@H]1O	13551284	LRUBQXAKGXQBHA-ZXXMMSQZSA-N	146.142	-2.000	0	50	AAAA	in-vitro
O=c1[nH]c(=O)[nH]c(=O)[nH]1	18044615	ZFSLODLOARCGLH-UHFFFAOYSA-N	129.075	-2.249	0	50	AAAA	biogenic,in-man
NC(=O)CCNC(=O)CCC(N)=O	57218621	XCAGWKCVVGEHBU-UHFFFAOYSA-N	187.199	-1.757	0	50	AAAA	in-vitro
COC(=O)[C@](C)(O)[C@@](C)(O)C(=O)O	1775956576	OHDPZUYCRQDXFK-BQBZGAKWSA-N	192.167	-1.254	0	50	AAAA	
NCC(=O)CN	1558643	RSBQFKJVGZVGFM-UHFFFAOYSA-N	88.110	-1.527	0	50	AAAA	in-vitro
NC[C@H](O)[C@H](O)CN	2384201	HYXRUUHQXNAFAJ-ZXZARUISSA-N	120.152	-2.374	0	50	AAAA	
NS(=O)(=O)NC[C@@H]1CCCNC1	19503746	SSEIPSQLNISNJP-ZCFIWIBFSA-N	193.272	-1.221	0	50	AAAA	in-vitro
OC[C@@H]1NC[C@H]2CNC[C@H]21	197586702	HQMJWVCANIVKHA-QYNIQEEDSA-N	142.202	-1.214	0	50	AAAA	
O=C(O)CN[C@H]1C=CS(=O)(=O)C1	4298576	SKCKHIKDKBWSEK-YFKPBYRVSA-N	191.208	-1.029	0	50	AAAA	in-vitro
O[C@@]1(c2nn[nH]n2)CCNC1	711217083	QMMJWKPNLAIMGD-YFKPBYRVSA-N	155.161	-1.619	0	50	AAAA	
N=C(N)Nc1nc(N)nc(N)n1	6523906	QLVPICNVQBBOQP-UHFFFAOYSA-N	168.164	-1.659	0	50	AAAA	in-vitro
COC(=O)[C@H](N)CN	19366965	WLXXVJOVQCQAMX-GSVOUGTGSA-N	118.136	-1.555	0	50	AAAA	
CO[C@H]1OC[C@H](O)[C@H](O)[C@H]1O	1041345	ZBDGHWFPLXXWRD-AZGQCCRYSA-N	164.157	-1.928	0	50	AAAA	in-vitro
NC(=O)[C@H](N)[C@@H]1CCOC1	37870616	MJXBRSNXTFSIDJ-RFZPGFLSSA-N	144.174	-1.165	0	50	AAAA	in-vitro
N[C@H]1[C@H](O)O[C@H](CO)[C@@H](O)[C@@H]1O	3860468	MSWZFWKMSRAUBD-QZABAPFNSA-N	179.172	-3.255	0	50	AAAA	endogenous,world
O=C(O)[C@H](O)[C@H](F)C(=O)O	39125626	FJOQIMHGSNFURE-NHYDCYSISA-N	152.077	-1.145	0	50	AAAA	
NC(=O)[C@H](O)[C@H]1CCOC1	50976959	SZSBDIGYBBYHJX-CRCLSJGQSA-N	145.158	-1.131	0	50	AAAA	in-vitro
CN1C(=O)[C@@H]2CNCCN2C1=O	72314731	JFKPCHOCFGJEHG-YFKPBYRVSA-N	169.184	-1.148	0	50	AAAA	in-vitro

AAAA - Excel

검색

파일 홈 삽입 페이지 레이아웃 수식 데이터 검토 보기 도움말

공유

붙여넣기 잘라내기 복사 서식 복사 | 클립보드

맑은 고딕 11 | 글꼴

자동 줄 바꿈 병합하고 가운데 맞춤 | 맞춤

일반 | 표시 형식

조건부 서식 표 서식 표준 나쁨 보통 좋음 경고문 계산 | 스타일

삽입 삭제 서식 | 셀

자동 합계 채우기 지우기 정렬 및 필터 찾기 및 선택 | 편집

추가 기능 | 추가 기능

H4 fx AAAA

	A	B	C	D	E	F	G	H	I
1	smiles	zinc_id	inchikey	mwt	logp	reactive	purchasab	tranche_na	features
2	CO[C@H]	4371221	ZBDGHWF	164.157	-1.928	0	50	AAAA	
3	C[S@@](=	34310585	TTXPTDCY	121.161	-1.15	0	50	AAAA	
4	NC(=O)N[	1843030	POJWUDA	158.117	-2.18	0	50	AAAA	endogenous,world
5	O=c1[nH]	1847417	YFQOVSG	144.086	-1.526	0	50	AAAA	biogenic,in-vitro
6	NC(=O)CI	9256947	JWRVCVW	157.217	-1.105	0	50	AAAA	in-vitro
7	CNC(=O)c	19844301	AFLLHPQF	141.134	-1.254	0	50	AAAA	
8	NCc1cc(=	26507110	UMEHPGX	141.13	-1.066	0	50	AAAA	in-vitro
9	CS(=O)(=	63014624	ZQJQZDH	152.219	-1.116	0	50	AAAA	
10	NC(=O)[C	2.43E+08	TTWXSENI	154.169	-1.248	0	50	AAAA	
11	N[C@@H]	1.05E+08	SNJHTJUW	148.162	-2.586	0	50	AAAA	
12	CNS(=O)(	2.38E+08	DRWMVYS	194.256	-1.476	0	50	AAAA	
13	O=c1[nH]	55860	SIVJYKAW	197.194	-1.1	0	50	AAAA	in-vitro
14	NCCOCCC	1651868	IWBOPFCI	148.206	-1.063	0	50	AAAA	in-vitro
15	CS(=O)(=	44594710	MASSUGV	187.242	-1.681	0	50	AAAA	in-vitro
16	NS(=O)(=	1.61E+08	DRBVAWC	180.229	-1.042	0	50	AAAA	
17	CN1C(=O)	1532166	MJJCZWP	128.087	-1.305	0	50	AAAA	biogenic,in-vitro
18	COC(=O)[	32098738	XWBUDPX	146.146	-1.638	0	50	AAAA	biogenic,in-vitro
19	NCCS(=O)	38251012	XZILNQFH	153.203	-1.648	0	50	AAAA	
20	O=C1N[C	2251996	PKNZKKFF	186.171	-2.031	0	50	AAAA	biogenic
21	O=C1O[C	71786849	QPJRIFFWI	174.152	-1.842	0	50	AAAA	
22	O=C(O)C(	1532571	WPAMZTV	178.14	-2.256	0	50	AAAA	in-vitro,metabolite
23	CC(=O)NC	35100900	GSMTYNS	174.2	-1.379	0	50	AAAA	
24	Cn1nc(CN	39064903	WMUKXU	154.173	-1.022	0	50	AAAA	
25	CNS(=O)(	1.6E+08	BLCRPDV	167.234	-1.611	0	50	AAAA	
26	O=C(O)C1	3.07E+08	IWHRKKRI	156.185	-1.12	0	50	AAAA	
27	NCCc1nnc	4.09E+08	PTNKDQU	179.183	-1.081	0	50	AAAA	
28	O=C(CO)[	902219	ZAQJHHRI	150.13	-2.738	0	50	AAAA	biogenic,in-vivo
29	Cn1ncc2c(	16969122	HAFKXNV	180.171	-1.058	0	50	AAAA	
30	O=C1NCC	4204973	QCMGTM	184.117	-1.043	0	50	AAAA	
31	NNc1n[nH	12958520	RDIMQHB	146.179	-1.06	0	50	AAAA	in-vitro
32	Nc1nc(N)r	16889989	KMZCSSC	166.148	-1.734	0	50	AAAA	
33	N[C@@H]	34413647	ZCZJKVCI	102.137	-1.722	0	50	AAAA	
34	COC[C@H	34426550	LNPOMYD	104.153	-1.081	0	50	AAAA	in-vitro

AAAA

준비 접근성: 사용할 수 없음 100%

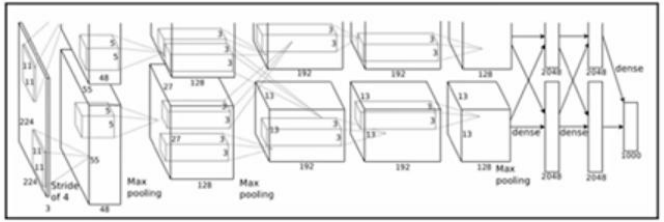
2009년도 알고리즘 수업

- Data structure (자료구조)
  - Python 기본 자료 구조: list, dictionary, set, tuple
  - Tree, Stack, Queue
- 알고리즘의 효율성을 표현하는 방법
  - 데이터 개수가 많을 수록 당연히 계산은 더 오래 걸림.
  - 데이터 개수에 비례해서 계산 시간이 얼마나 늘어나는지를 표시
  - Big O notation: O(n)
- 8억개 물질 중에서 원하는 구조를 추출하는 방법
  - Sorting, Searching, Graph

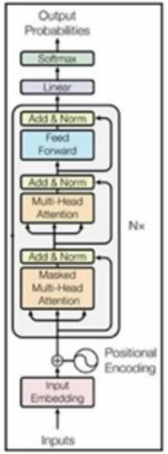
Youtube (Y combinator) Andrej Karpathy: Software Is Changing (Again)
https://youtu.be/LCEmiRjPEtQ?si=O8syc4Of6gXuShnb

Software 1.0	Software 2.0	Software 3.0
computer code	weights	prompts
↓ programs	↓ programs	↓ programs
computer	neural net	LLM

became programmable in **~1940s**



fixed function neural net
e.g. AlexNet: for image recognition (**~2012**)



~2019

LLM = programmable neural net!

Youtube (Y combinator) Andrej Karpathy: Software Is Changing (Again)
https://youtu.be/LCEmiRjPEtQ?si=O8syc4Of6gXuShnb

Example: Sentiment Classification

Software 1.0

python Copy

```python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?;:")          # crude token clean-up
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
```

Software 2.0

10,000 positive examples

10,000 negative examples

encoding (e.g. bag of words)

train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags
<REVIEW> … </REVIEW>, respond with **exactly one word**, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

Example 1
<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>
POSITIVE

Example 2
<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>
NEGATIVE

Example 3
<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>
NEGATIVE

Now classify the next review.